

A SMALL EQUALITY-CASE NOTE FOR THE POLAR SCHNEIDER COLLECTION INEQUALITY

ARXIV_AUTOMATIC_MATH RUN FA_BANACH.001

SOURCE AND STATUS

This note concerns Julian Haddad, Dylan Langharst, Galyna V. Livshyts, and Eli Putterman, *On the polar of Schneider's difference body*, arXiv:2503.06191.

Status: `partial_result_likely_valid`.

The source paper proves a polar Schneider inequality for a collection $\mathcal{K} = (K_0, \dots, K_m)$ of convex bodies. In Theorem 11 it states that a necessary condition for equality is that all K_i are centered ellipsoids, while a sufficient condition is that all K_i are the same centered ellipsoid. The authors then note that they do not know whether equality can hold when all K_i are centered ellipsoids but at least one of them differs from the others.

The present packet closes only the case $m = 1$.

DEFINITIONS

For two convex bodies $K_0, K_1 \subset \mathbb{R}^n$, the source paper's collection difference body is

$$D^1(K_0, K_1) = \{x \in \mathbb{R}^n : K_0 \cap (K_1 + x) \neq \emptyset\} = K_0 + (-K_1).$$

If the bodies are centered ellipsoids, they are centrally symmetric, so $-K_1 = K_1$. The $m = 1$ instance of Theorem 11 is

$$|D^{1,\circ}(K_0, K_1)| |K_1| \leq |B_2^n| |D^{1,\circ}(B_2^n, B_2^n)|.$$

Since $D^1(B_2^n, B_2^n) = 2B_2^n$, the right-hand side is

$$|B_2^n| |(2B_2^n)^\circ| = 2^{-n} |B_2^n|^2.$$

IDEA OF THE PROOF

When $m = 1$, the collection body is just a Minkowski sum. If E_0, E_1 are centered ellipsoids and their polar volumes agree, then their ordinary volumes also agree. Brunn–Minkowski says $E_0 + E_1$ has volume at least the volume of $2E_1$, with equality only when the two ellipsoids are homothetic. Since the volumes and centers agree, homothety means equality of the ellipsoids. Applying Blaschke–Santaló to $E_0 + E_1$ then gives the desired strictness unless $E_0 = E_1$.

Theorem 1. *Let $E_0, E_1 \subset \mathbb{R}^n$ be centered ellipsoids satisfying*

$$|E_0^\circ| = |E_1^\circ|.$$

Then equality in the $m = 1$ case of the source paper's Theorem 11 holds if and only if $E_0 = E_1$. In particular, if $E_0 \neq E_1$, then

$$|(E_0 + E_1)^\circ| |E_1| < 2^{-n} |B_2^n|^2.$$

Proof. For a centered ellipsoid $E = AB_2^n$,

$$|E||E^\circ| = |B_2^n|^2.$$

Thus the assumption $|E_0^\circ| = |E_1^\circ|$ implies

$$|E_0| = |E_1| =: V.$$

Set $L = E_0 + E_1$. The body L is centrally symmetric, so the Blaschke–Santaló inequality applies with the origin as Santaló point:

$$|L^\circ||L| \leq |B_2^n|^2.$$

On the other hand, Brunn–Minkowski gives

$$|L|^{1/n} = |E_0 + E_1|^{1/n} \geq |E_0|^{1/n} + |E_1|^{1/n} = 2V^{1/n}.$$

Therefore $|L| \geq 2^n V$, and hence

$$|(E_0 + E_1)^\circ||E_1| = |L^\circ|V \leq \frac{|B_2^n|^2 V}{|L|} \leq 2^{-n} |B_2^n|^2.$$

This is exactly the $m = 1$ collection inequality for centered ellipsoids.

It remains to identify equality. Equality in the chain above requires equality in Brunn–Minkowski. For full-dimensional convex bodies this forces E_0 and E_1 to be homothetic. Their volumes are equal, so the homothety ratio is one; since both ellipsoids are centered at the origin, this gives $E_0 = E_1$. Conversely, if $E_0 = E_1 = E$, then

$$|(2E)^\circ||E| = 2^{-n}|E^\circ||E| = 2^{-n}|B_2^n|^2,$$

so equality holds. This proves the claim. $\square$

LIMITATIONS

This packet does not settle the source paper’s equality question for $m \geq 2$. There the collection body is no longer just a sum of two full-dimensional bodies in $\mathbb{R}^n$, and the above Brunn–Minkowski equality argument does not directly apply.

It also does not address Schneider’s original higher-order conjecture on the minimum of $|D^m(K)||K|^{-m}$, which remains a hard Petty-adjacent convex geometry problem.

NOVELTY CHECK

Local run indexes were searched for arXiv id **2503.06191**, the title, “Polar Schneider”, “Schneider difference body”, “Theorem 11”, and “same centered ellipsoid”. No prior packet for this equality subcase was found. A bounded web search on 2026-07-03 for the exact title together with phrases from the equality-gap sentence surfaced the source arXiv page and no separate answer to this $m = 1$ equality subcase.

Novelty confidence is modest: the observation is elementary and may be known to experts, but it is not stated in the source paper’s equality discussion.

HUMAN REVIEW RECOMMENDATION

Review as a narrow likely-valid partial result. The main point to verify is that the equality characterization in Brunn–Minkowski is being applied in its standard full-dimensional form.

REFERENCES

- [1] J. Haddad, D. Langharst, G. V. Livshyts, and E. Putterman, *On the polar of Schneider's difference body*, arXiv:2503.06191, 2025.
- [2] R. J. Gardner, *The Brunn–Minkowski inequality*, Bull. Amer. Math. Soc. (N.S.) **39** (2002), 355–405.
- [3] M. Fradelizi, M. Meyer, and A. Zvavitch, *Volume product*, in *Harmonic analysis and convexity*, Adv. Anal. Geom. 9, De Gruyter, 2023, 163–222.